

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

Lecture 2 Animations: Student Guide

KKKC6353 Nanoelectronic Devices — learn by experimenting

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How to use this guide

Seven interactive animations go with the Lecture 2 notes. They are at **faizverse.github.io/nanoelectronic-devices/L2** and work on a laptop or phone, online or saved for offline use.

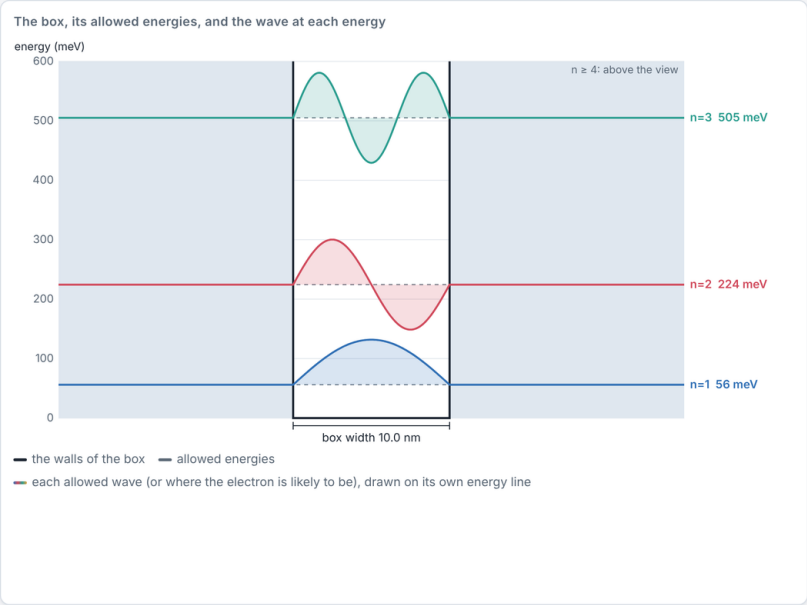
For each animation:

1. **Read first.** Skim the notes section listed, then read *What it shows* below. The page opens in a simple view; press **More detail** when you want the numbers and graphs.
2. **Predict, try, explain.** Before you move a slider, write down what you expect. Then try it. If the result surprised you, work out why.
3. **Answer the questions** at the end of each page in this guide, then check your answers under *Check yourself* on the web page.
4. **Note the big idea** in one sentence of your own.

Allow about 10 minutes per animation. Work through them in order: each builds on the one before.

#	Animation	Read first	Link
1	Particle in a box	Doc 1, § 6	particle-in-a-box
2	Density of states, 3D to 0D	Doc 2, § 2	density-of-states
3	Quantum-well emission	Doc 2, § 3	quantum-well-emission
4	Quantum cascade laser	Doc 2, § 4	quantum-cascade-laser
5	Carbon nanotubes: roll your own	Doc 2, § 6	carbon-nanotubes
6	Quantum-dot colour	Doc 2, § 7	quantum-dot-colour
7	Growing quantum dots	Doc 3, § 3–4	quantum-dot-growth

1 Particle in a box



Link: https://faizverse.github.io/nanoelectronic-devices/L2/particle-in-a-box.html
Read first: Document 1, Section 6

What it shows. Trap an electron in a tiny box and only certain energies are allowed, like the notes of a guitar string. Squeeze the box, make the walls leaky, and watch the allowed energies and the waves change.

Only certain notes fit. A guitar string pinned at both ends can only vibrate with a whole number of humps. An electron trapped in a box is the same: its wave must fit inside the box, so only whole numbers of half-waves are allowed. Each one has its own energy, and no energies in between are possible. The lowest is labelled n = 1, the next n = 2, and so on.

Smaller box, bigger energies. Squeeze the box and every wave must wiggle more tightly, so every energy goes up: halve the width and the energies go up four times. A lighter particle also has more widely spaced energies.

It never sits still. There is no n = 0, so the lowest energy is never zero.

Watch it move. In “the wave” view each wave vibrates up and down like a plucked string, and higher energies vibrate faster. In the “where it is likely” view nothing moves: the chance of finding the electron at each place stays the same. The energy scale and the 40 nm window are fixed, so as you move the box width you see the box itself narrow or widen and the energies climb or fall.

Real walls leak. Choose “Real (leaky)” walls and the wave spills a little way into the walls. The energies drop (a 10 nm GaAs box goes from 56 to 32 meV), and a shallow box can only hold a few of them.

> **The big idea.** A trapped electron can only have certain energies, like the notes of a guitar string. The smaller the box, the further apart they are, and the lowest is never zero.

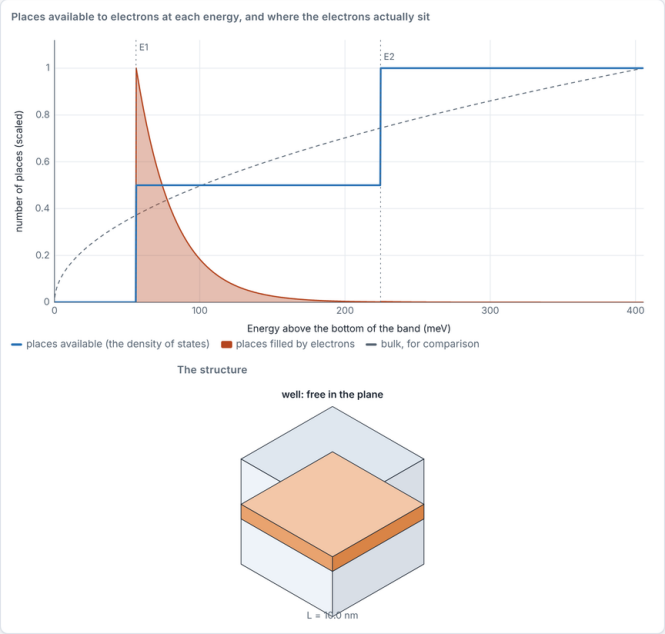
Try this

1. Perfect walls, 10 nm box. Read the lowest energy. **Predict** what it will be at 5 nm, then check.
2. Switch to “where it is likely”. In the n = 2 state, where is the electron never found?
3. Switch to real, leaky walls. Do the energies go up or down? Look at the waves near the walls.
4. Leaky walls, about 2 nm. How many allowed energies are left?

Questions (check your answers on the page, under *Check yourself*)

- Why can a trapped electron not have zero energy?
- Why do the energies go down when the walls leak?
- Two quantum dots of the same material glow red and blue. Which is smaller?

2 Density of states, 3D to 0D



Link: https://faizverse.github.io/nanoelectronic-devices/L2/density-of-states.html
Read first: Document 2, Section 2

What it shows. How many places are there for electrons at each energy? Trap the electrons in one, two or three directions and watch the places bunch up, from a smooth curve to a staircase, sharp peaks and finally single sharp levels.

Places to sit. Think of the electrons as an audience filling a theatre with seats on many levels, one level for each energy. The blue curve is the number of seats on each level: the **density of states**. The rust area shows where the electrons actually sit. They fill the lowest levels first, and warmth spreads them a little higher.

Trapping reshapes the seats. In bulk material there are few places at the very bottom, and more as the energy rises. Trap the electrons in one direction (a quantum well) and the places come in a **staircase**. Trap them in two (a wire) and they bunch into **sharp peaks**. Trap them in all three (a dot) and only **single sharp levels** remain, like an atom.

Why it matters. The more directions are trapped, the narrower the range of energies the electrons occupy. A laser wants all its electrons at one energy, which is why quantum-well lasers need less current than bulk lasers, and why quantum-dot lasers are better still.

> **The big idea.** Trapping electrons in more directions bunches the places available to them: a smooth curve, a staircase, sharp peaks, then single sharp levels.

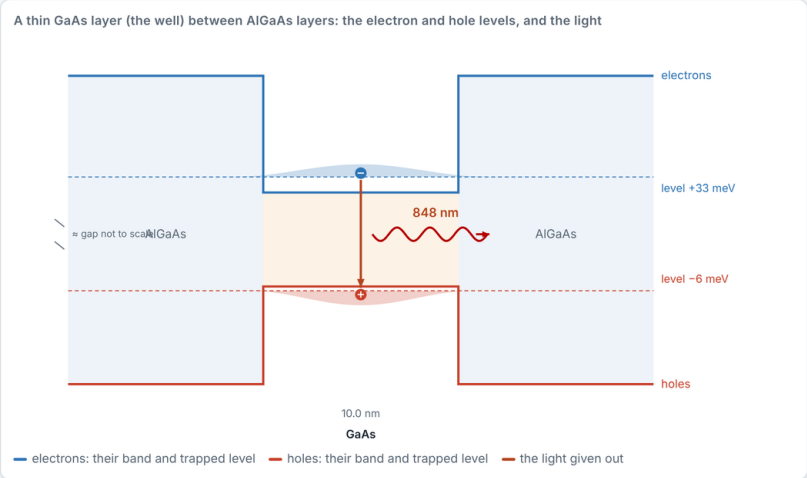
Try this

1. Bulk: see how widely the electrons (rust) are spread in energy. **Predict** what happens in a well.
2. Well: where is the first step, and where do the electrons sit?
3. Wire, then Dot: do the electrons occupy a wider or a narrower range of energies?
4. Make the trap smaller and cool it down. When do almost all the electrons sit in the lowest level?

Questions (check your answers on the page, under *Check yourself*)

- Is a quantum dot one-dimensional?
- Why does a quantum-well laser need less current than a bulk laser?
- Real quantum-dot lasers contain millions of dots. Why are their levels not perfectly sharp?

3 Quantum-well emission



Link: https://faizverse.github.io/nanoelectronic-devices/L2/quantum-well-emission.html
Read first: Document 2, Section 3

What it shows. In a quantum-well laser, electrons and holes meet in a thin layer and give out light. Change the thickness of the layer and watch the colour of the light change, without changing the material.

A trap for both. A thin layer of GaAs sits between layers of AlGaAs. Electrons have a lower energy in the GaAs, and so do holes (the empty places left where electrons are missing). So the thin layer is a well that traps both.

They meet and give out light. An electron drops into a hole and gives out its energy as light. Because each is trapped in a box, each sits on a trapped level a little away from the bottom of its band. So the light carries slightly more energy than GaAs alone would give.

Thinner layer, shorter wavelength. Make the well thinner and both trapped levels move further out, so the light has more energy and a shorter wavelength. The designer tunes the colour by choosing the thickness. GaAs wells give near-infrared light, used in optical links, sensors and laser printers.

> **The big idea.** Electrons and holes trapped in a thin layer give out light with a little more energy than the material alone, so a thinner layer gives a shorter wavelength.

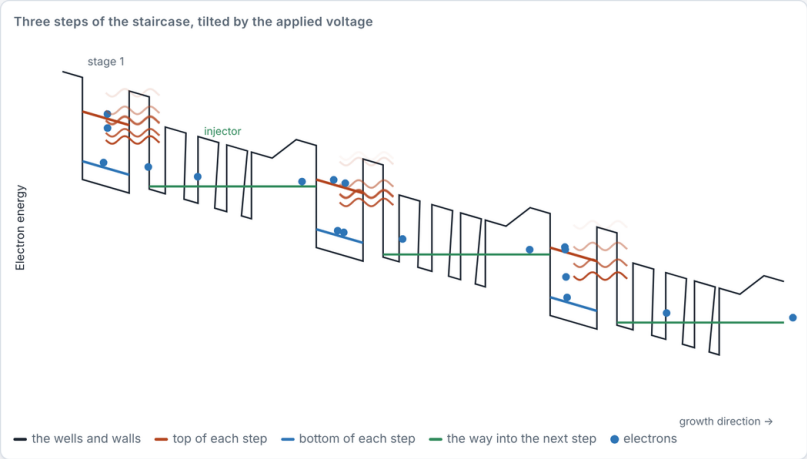
Try this

1. 10 nm: read the wavelength and compare it with bulk GaAs (871 nm).
2. **Predict**: will a 5 nm well give a longer or a shorter wavelength? Check.
3. As you make the well thinner, which level moves more, the electron's or the hole's?
4. Find the thickness that gives 830 nm.

Questions (check your answers on the page, under *Check yourself*)

- Why is the light from a quantum well bluer than from bulk GaAs?
- Why does the hole's level move less than the electron's?
- How can one material make lasers of different colours?

4 Quantum cascade laser



Link: https://faizverse.github.io/nanoelectronic-devices/L2/quantum-cascade-laser.html
Read first: Document 2, Section 4

What it shows. Electrons run down a staircase of quantum wells, giving out a photon at every step. Turn up the voltage until the steps line up and the laser switches on, and change the step size to change the colour.

A staircase instead of a cliff. In an ordinary laser an electron makes one big drop, meets a hole, gives out one photon and is gone. A quantum cascade laser uses only electrons. It is a stack of many thin quantum wells, and the applied voltage tilts the stack into a staircase.

A photon at every step. At each step the electron drops from a higher trapped level (rust) to a lower one (blue) and gives out a photon. It is not destroyed: it moves on through a thin wall, by tunnelling, into the top of the next step, and drops again. One electron can give out a photon at every step.

The steps must line up. With too little voltage the way into the next step (green) sits too low; with too much it sits too high. Only when it lines up with the top of the next step do electrons flow and the laser switch on. The step size, set by the thickness of the wells, sets the colour: infrared light, used for sensing tiny traces of gases.

> **The big idea.** An electron runs down a staircase of wells and gives out a photon at every step, but only when the voltage lines the steps up.

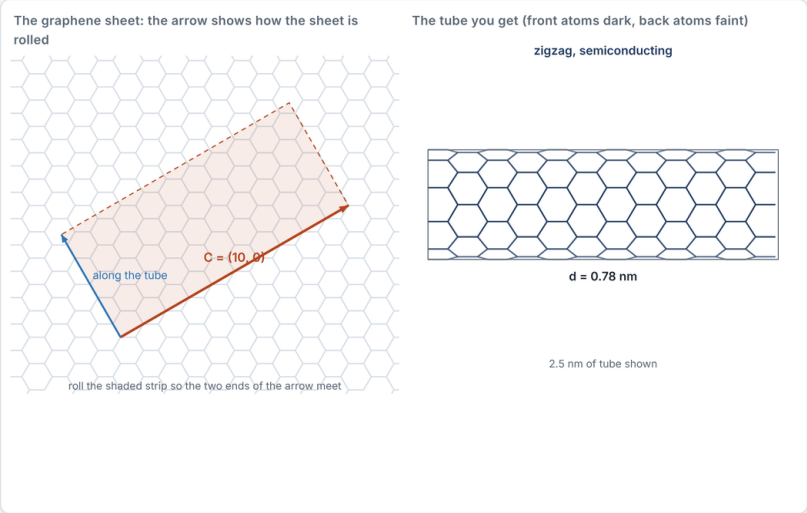
Try this

1. Raise the voltage slowly from zero. At what voltage does the laser switch on?
2. Keep going. What happens when the steps pass each other?
3. **Predict**: does a wider active well give a longer or a shorter wavelength? Check, then find the new voltage that lines the steps up.

Questions (check your answers on the page, under *Check yourself*)

- Why is a quantum cascade laser called unipolar?
- What sets the colour of its light?
- Does “one electron, many photons” make it very powerful?

5 Carbon nanotubes: roll your own



Link: https://faizverse.github.io/nanoelectronic-devices/L2/carbon-nanotubes.html
Read first: Document 2, Section 6

What it shows. Roll a sheet of graphene into a tube and the way you roll it decides whether the tube is a metal or a semiconductor. Choose the roll, see the tube, and find out whether it could make a transistor.

Rolling up graphene. Graphene is a single sheet of carbon atoms in hexagons, like chicken wire. Roll it into a seamless tube about a nanometre across and you have a carbon nanotube. The arrow on the sheet shows which two points are joined when it is rolled. Two whole numbers, (n, m), describe the arrow, and so the tube.

Metal or semiconductor. How the sheet is rolled decides how the electron waves fit around the tube. If n − m can be divided by 3, the tube conducts like a metal. Otherwise it is

a semiconductor, and the thinner the tube, the bigger its band gap. About one tube in three is metallic.

Why it matters. A transistor must be able to switch off, which needs a semiconducting tube. A metallic tube cannot be switched off and would short the transistor. Every growth method makes a mix, so tubes have to be sorted (Document 3, Section 5).

> **The big idea.** How graphene is rolled, (n, m), decides whether a nanotube is a metal or a semiconductor: if n − m divides by 3 it is a metal.

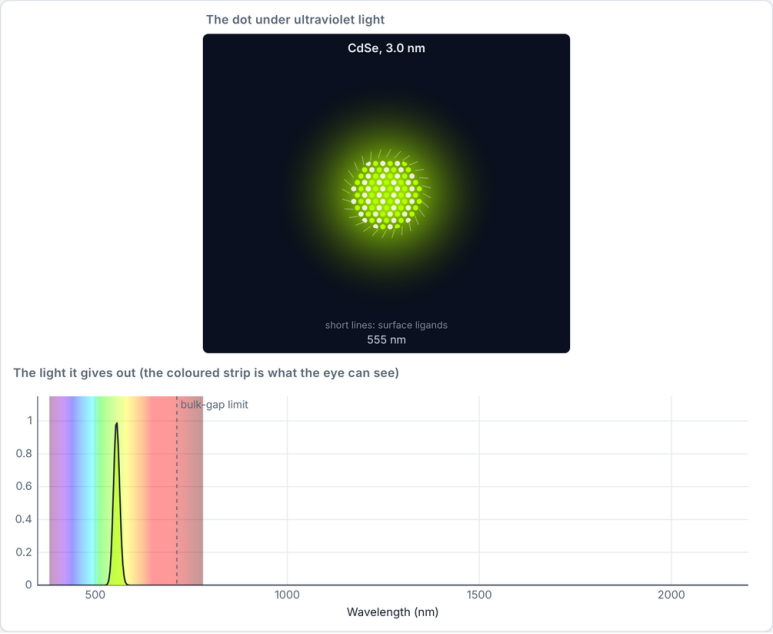
Try this

1. (10, 0): **predict** metal or semiconductor from n − m, then check. Now try (9, 0).
2. (10, 10), armchair: metal or semiconductor? Explain it from the rule.
3. Press Random ten times. How many tubes could make a working transistor? Is it close to two in three?

Questions (check your answers on the page, under *Check yourself*)

- What decides whether a nanotube is a metal or a semiconductor?
- Does a perfect metallic nanotube have zero resistance?
- Why are nanotube chips so hard to make?

6 Quantum-dot colour



Link: https://faizverse.github.io/nanoelectronic-devices/L2/quantum-dot-colour.html
Read first: Document 2, Section 7

What it shows. Dots of the same material glow different colours depending only on their size. Grow the dot, change the material, and watch the colour of its light change.

Size sets the colour. When a dot absorbs ultraviolet light, an electron is lifted up and leaves a hole behind. The pair, called an exciton, then recombines and gives out light. In a small dot the electron and hole are squeezed tightly, like a particle in a small box, so their energies

are pushed up and the light is bluer. In a large dot they are squeezed less and the light is redder.

The material sets the range. Each material has its own band gap, and no dot can glow below it. CdSe dots cover the visible colours. PbS dots of the same size glow in the infrared, which the eye cannot see.

Why it matters. Dots give pure colours in a narrow band, set by their size. That is why they are used in "QLED" and "QD-OLED" television screens, and to tag cells in different colours under a microscope.

> **The big idea.** Within one material, smaller dots glow bluer and larger dots redder; the material decides the range of colours.

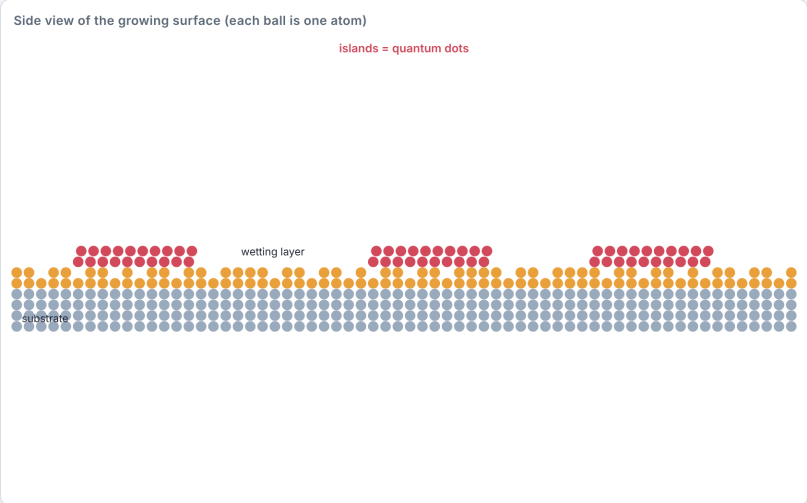
Try this

1. CdSe: slide from 2 nm to 7 nm and note the colours.
2. Which CdSe size gives the green of a screen, about 530 nm?
3. Change to PbS at 3 nm. Where has the light gone, and why?
4. Try InP, the cadmium-free material used in screens. Can it make the same colours as CdSe?

Questions (check your answers on the page, under *Check yourself*)

- Why does a smaller dot glow bluer?
- Why does a 3 nm PbS dot glow in the infrared when a 3 nm CdSe dot glows green?
- Why do display makers want all their dots the same size?

7 Growing quantum dots



Link: https://faizverse.github.io/nanoelectronic-devices/L2/quantum-dot-growth.html
Read first: Document 3, Sections 3–4

What it shows. Two ways quantum dots build themselves: a squeezed film that breaks up into islands, and tiny crystals that seed and grow in a hot flask. Add material, inject, watch the dots grow, and stop them at the colour you want.

Self-assembled dots. Atoms of InAs land on a GaAs crystal, one layer at a time. InAs atoms are naturally a little further apart (about 7%), so the film is squeezed to fit, like a carpet

slightly too big for its room. The first layer lies flat: the **wetting layer**. After about 1.7 layers of atoms the squeeze is too much, and new material bunches into small islands where it can spread out. The islands are quantum dots. No pattern is drawn: the dots build themselves.

Dots from a flask. Inject the ingredients into a hot liquid and a sudden burst of tiny seed crystals forms, all at the same moment. After that no new seeds form; the existing ones just grow. Because they all started together, they stay nearly the same size, and as they grow their colour moves from blue towards red. Cooling the flask ("Quench") stops the growth at the colour you want.

> **The big idea.** Dots can build themselves: a squeezed film breaks into islands after about 1.7 layers of atoms, and dots seeded together in a hot flask grow to nearly the same size.

Try this

1. On a crystal: add material slowly. When do islands appear?
2. Keep adding. Does the wetting layer get thicker after the islands appear?
3. In a hot flask: press Inject and watch the dots grow and change colour.
4. Press Quench when the dots glow green. Then inject again and stop at red. Which took longer?

Questions (check your answers on the page, under *Check yourself*)

- Is self-assembled (Stranski–Krastanov) growth top-down or bottom-up?
- Why do the flask-grown dots come out nearly the same size?
- How do you get red dots instead of green from the same recipe?